

Transforming & Integrating

Rubber, meet road.

(So, do we just *do the thing*?)

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Transforming & Integrating



The “last” “phase” is about choreographing change; this is inherently risky. Doing so in complex systems requires learningful approaches.

Transforming & Integrating



Remember: design learns by doing through experimentative approaches.

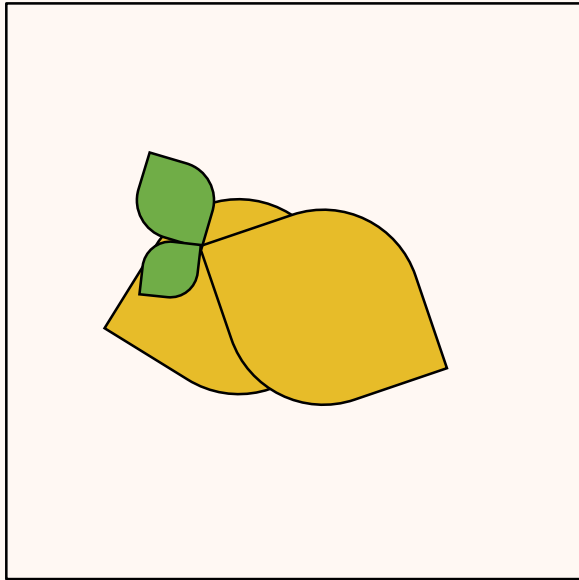
Lean Experimentation



...uses safe-to-fail experiments targeting critical hypotheses to drive rapid learning.

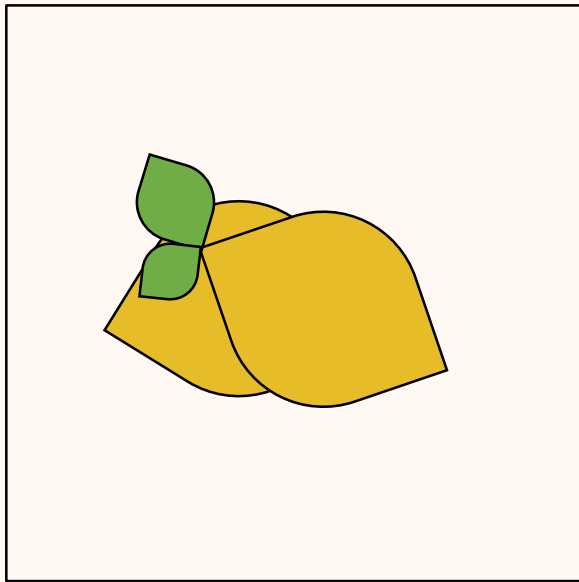
Lean Experimentation

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- 1. Analyze your design concept and identify assumptions.
- 2. Evaluate assumptions by impact and uncertainty.
- 3. Starting with high impact-high uncertainty assumptions, frame testable hypotheses.
- 4. Craft experiments to validate (or invalidate!) your hypothesis.

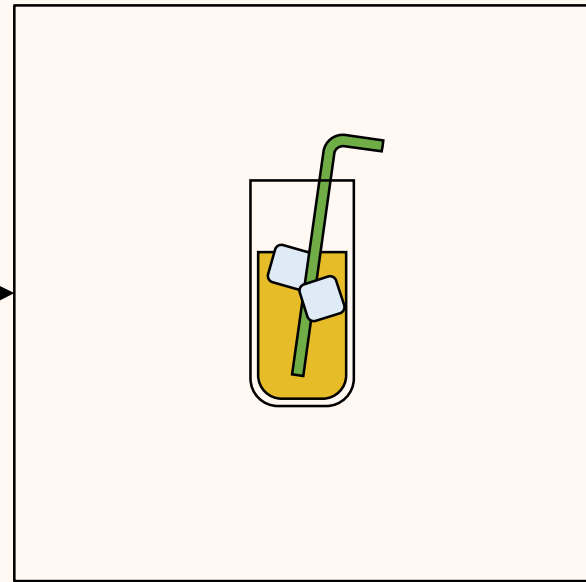
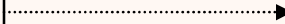


Current State

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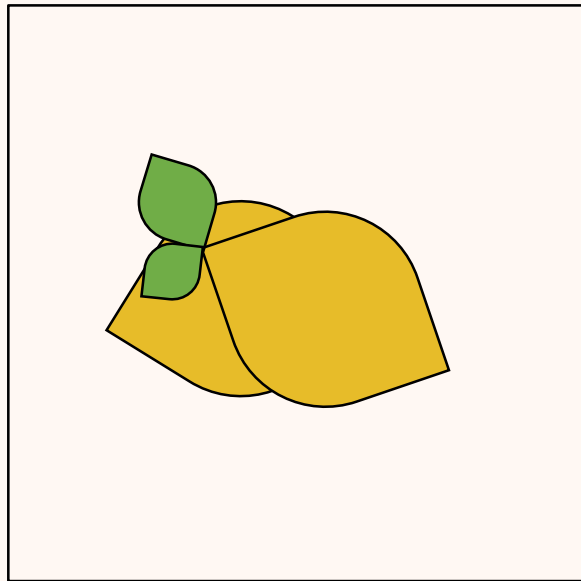


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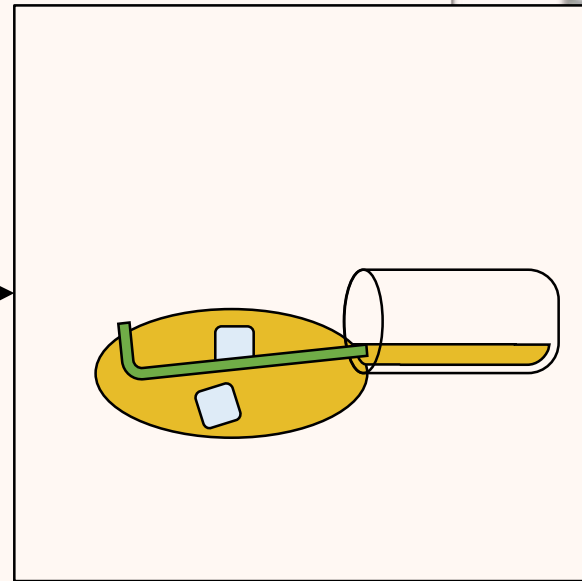
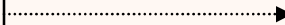


Alternative

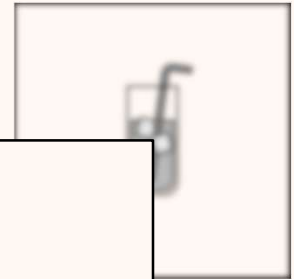
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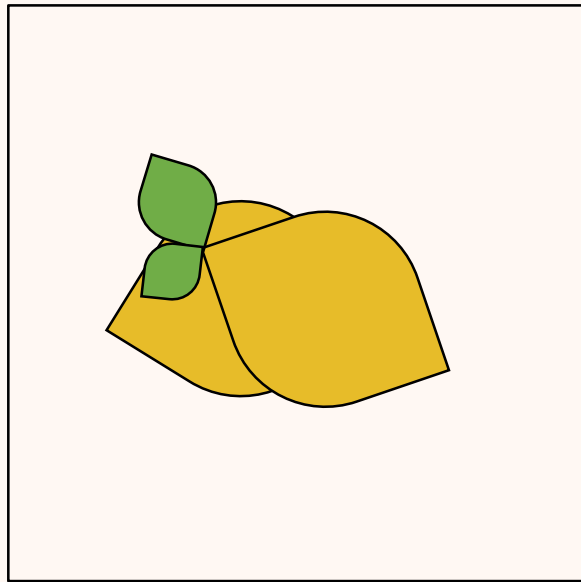
Current State



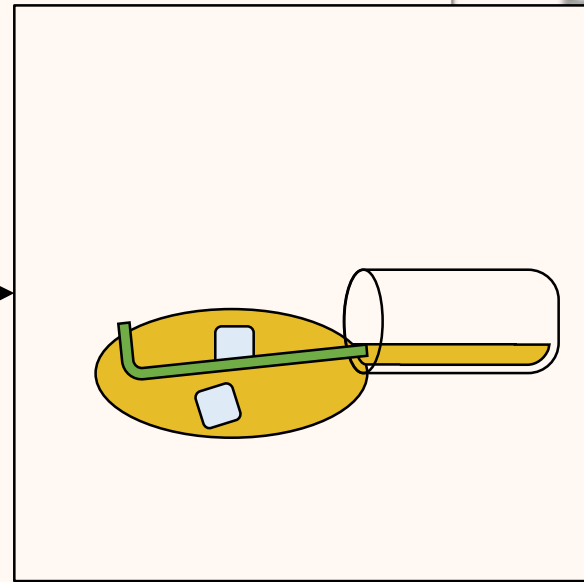
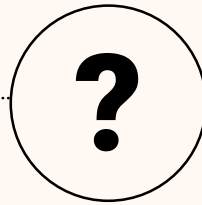
Oh no!



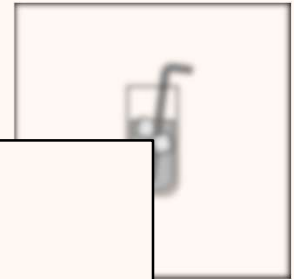
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Current State



Oh no!



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Impact

The importance of a particular assumption to our concept.

Ask: "If this assumption is incorrect, how and to what extent would our concept be effected?"

High Uncertainty
High Impact

Low Uncertainty
Low Impact

Uncertainty

Confidence in a given assumption.
Distinct from likelihood.



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Assumption formulation:

“Books will keep toddlers on the potty longer!”

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Hypothesis formulation:

“Giving a toddler a selection of 3 books while they use the potty will increase their overall potty duration by >30%.”

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Prototyping

Things as partners.

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Prototyping

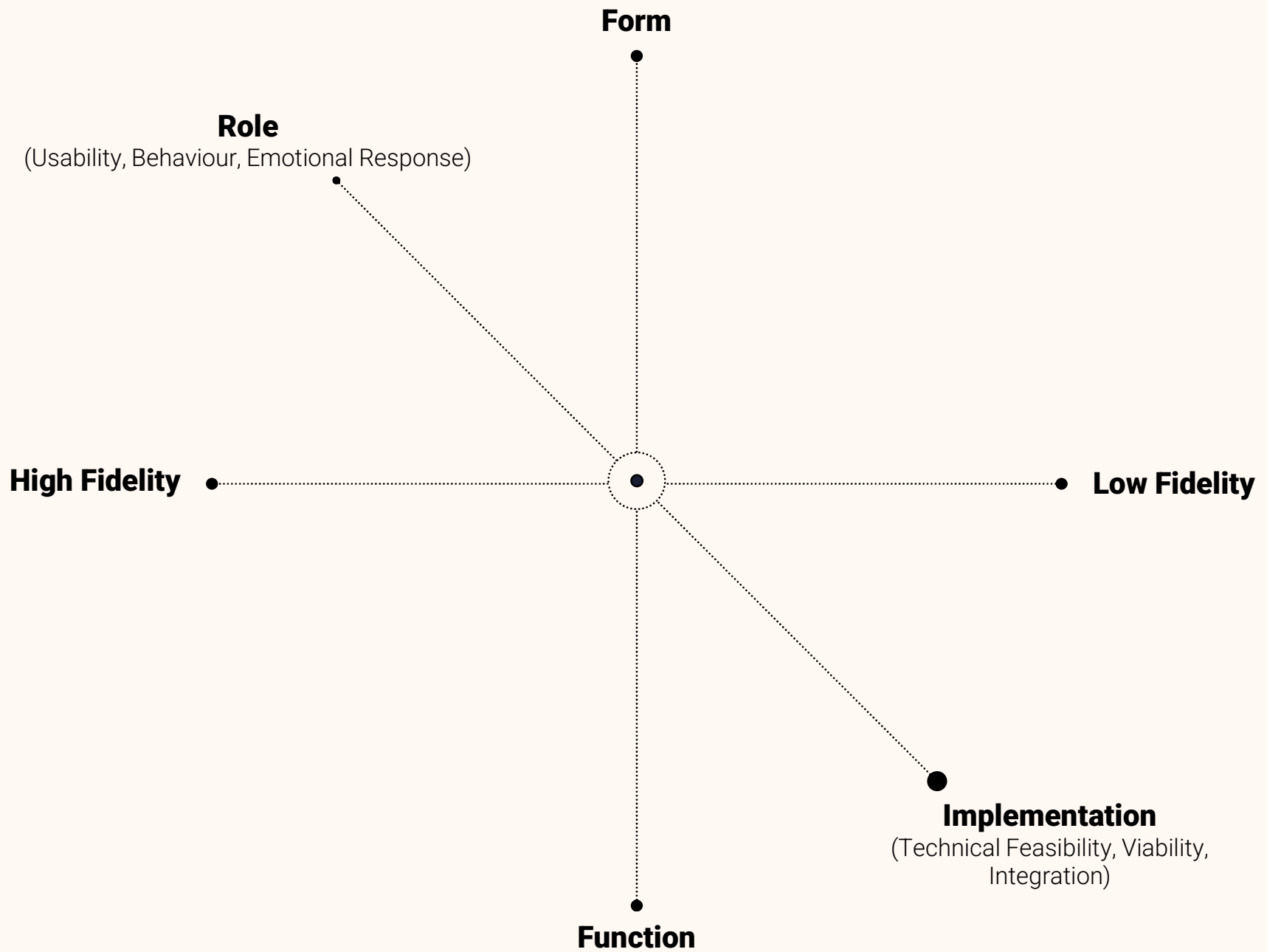


A prototype is a disposable, temporary stimuli that tests a hypothesis by eliciting a response. A vehicle for producing knowledge.

Prototyping



Prototypes can be used at any stage in the design process to ideate, prompt, test, even communicate and persuade.



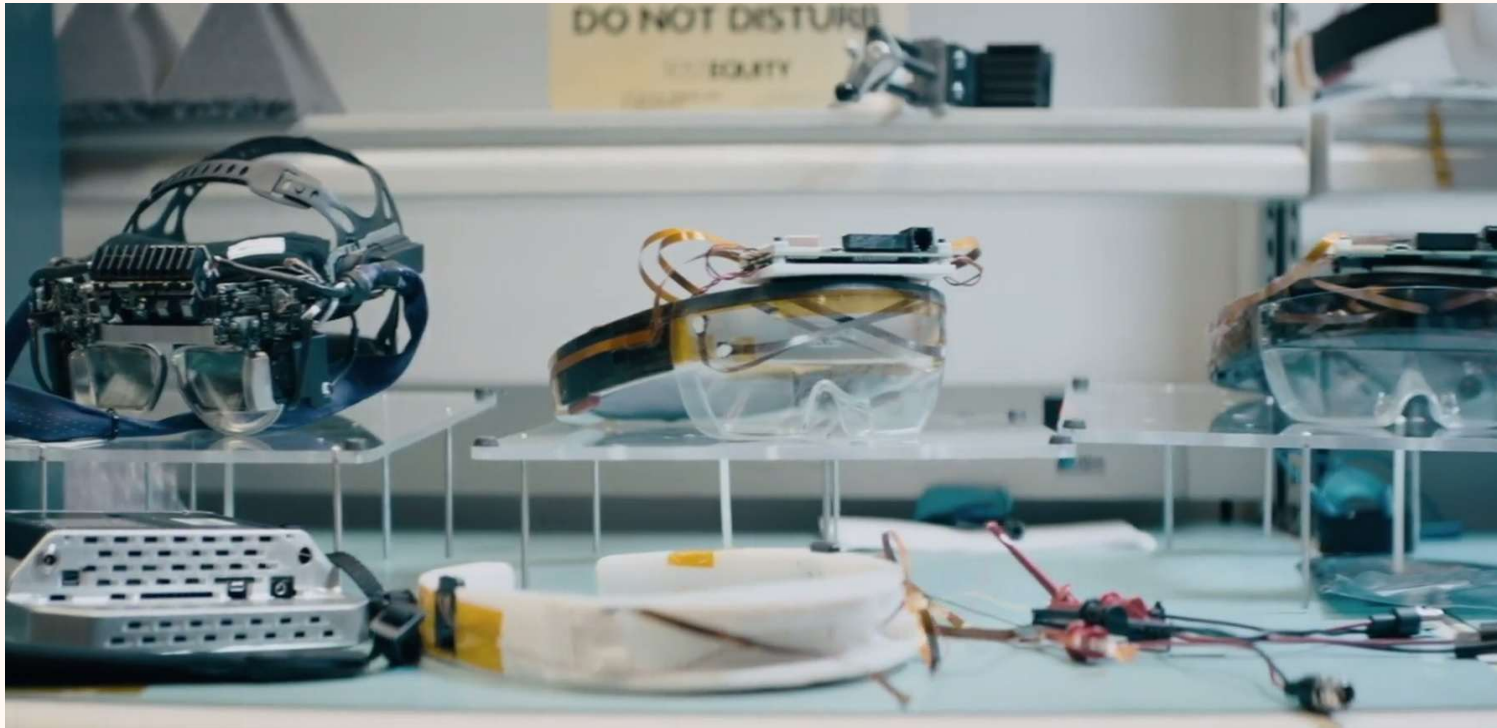
Roles



Roles

Roleplay is the most common means of prototyping roles.

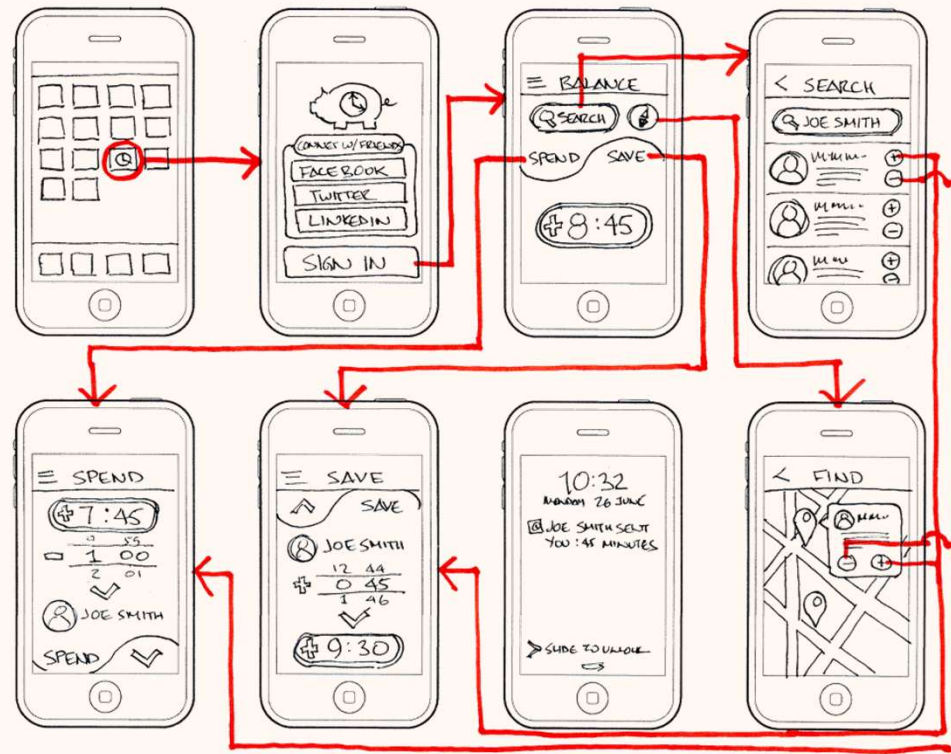
Roles **Objects**



Objects

Tangible prototypes are used to explore and test qualities of physical objects.

Roles Objects **Information**



Information

Wireframes and paper prototypes are used to test the usability, navigability, and learnability of information

Roles
Objects
Information
Services/Processes



Services/Processes

Storyboard and service blueprints are both low fidelity ways to prototype processes. Higher fidelity prototypes might involve a combination of tangible prototypes, role playing, etc.

Roles
Objects
Information
Services/Processes
Models

The Business Model Canvas

Designed for:

Designed by:

On:

Iteration:

Key Partners What are our Key Partners? Who are our key suppliers? Which Key Resources are we acquiring from partners? Which Key Activities do partners perform?	Key Activities What Key Activities do our Value Propositions require? Our Distribution Channels? Customer Relationships? Revenue Streams?	Value Propositions What value do we deliver to the customer? What bundle of products and services are we offering to each Customer Segment? Which customer needs are we satisfying?	Customer Relationships What type of relationship does each of our Customer Segments expect us to establish and maintain with them? Which ones have we established? How are they integrated with the rest of our business model? How costly are they?	Customer Segments For whom are we creating value? Who are our most important customers?
	Key Resources What Key Resources do our Value Propositions require? Our Distribution Channels? Customer Relationships? Revenue Streams?		Channels Through which Channels do our Customer Segments want to be reached? How are we reaching them now? How are our Channels integrated? Which ones work best? Which ones are most cost-efficient? How are we integrating them with customer routines?	
Cost Structure What are the most important costs inherent in our business model? Which Key Resources are most expensive? Which Key Activities are most expensive?			Revenue Streams For what value are our customers really willing to pay? For what do they currently pay? How are they currently paying? How would they prefer to pay? How much does each Revenue Stream contribute to overall revenues?	

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Model

Business or mission models, Theories of Change, and logical frameworks are instantiations of models. This class focuses on how our solution operates within, and interacts with, the larger context. We use approaches like Lean Experimentation to decompose models into testable components – validating demand, for example.

- We create **scenarios** to isolate the dimensions of the concept we want to use the prototype we want to explore.
- Many of the **same research methods** we employed at the outset are appropriate for capturing data, analyzing and synthesizing, and drawing out insights during testing.
- Testing is commonly **qualitative**, but frequently benefits from **quantitative measures**.

Exercise

In small groups:

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- 1. **Analyze the Women Lead in Emergencies Concept**
- 2. **Identify assumptions and plot on the Impact/Uncertainty 2x2**
- 3. **Develop a falsifiable hypothesis**
- 4. **How would you test?**

Problematizing Design



1. **Authority, objectivity, designer bias.**
2. **Who gets to be a *designer*? Expert-led vs. diffuse.**
3. **Coloniality and relationality**
4. **Is human centeredness systems blindness?**